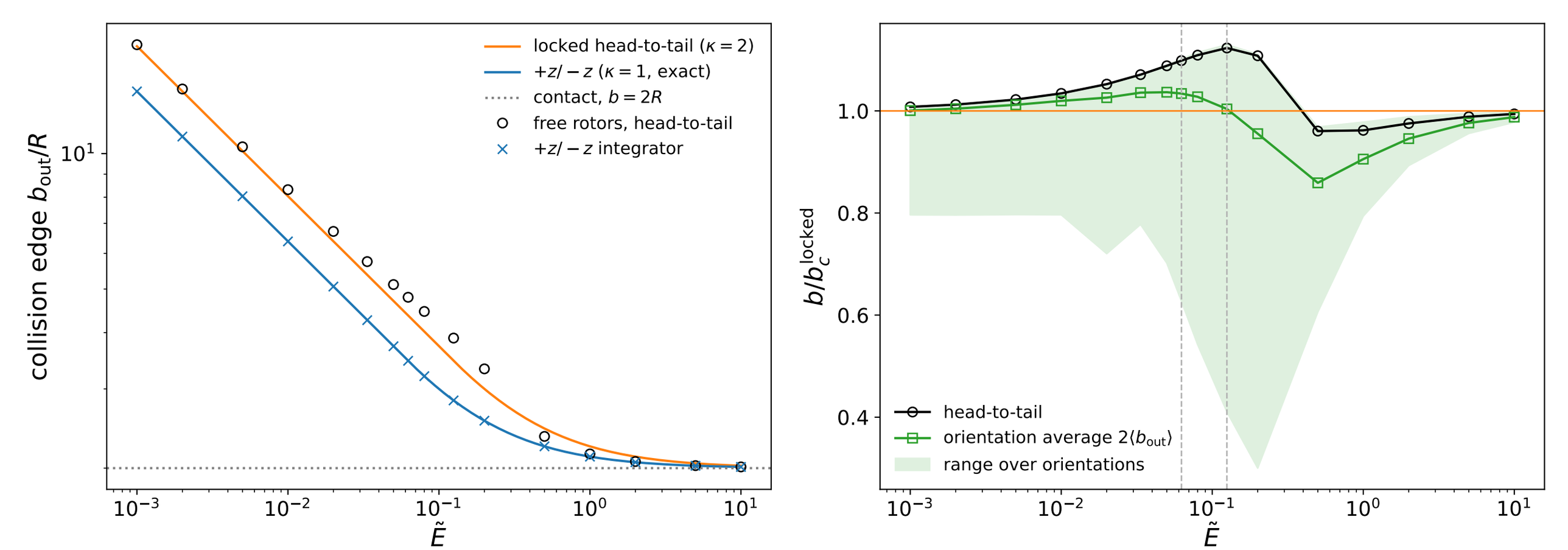
locked head-to-tail ($\kappa = 2$)
$+z/-z$ ($\kappa = 1$, exact)
contact, $b = 2R$
free rotors, head-to-tail
$+z/-z$ integrator
collision edge b_{out}/R
10^1
10^{-3}
10^{-2}
10^{-1}
10^0
10^1
$\tilde{E}$
b/b_c^{locked}
1.0
0.8
0.6
0.4
head-to-tail
orientation average $2\langle b_{\mathrm{out}}\rangle$
range over orientations
10^{-3}
10^{-2}
10^{-1}
10^0
10^1
$\tilde{E}$